

- 5 The gene *BRCA1* is expressed in breast tissue and in several other tissues of the body. *BRCA1* codes for a tumour suppressor protein. This protein is involved in either repairing damaged DNA, or in triggering the death of a cell if DNA cannot be repaired.

- (a) State **one** reason why a doctor would recommend that a person has genetic testing for a mutation in the gene *BRCA1* **and** explain why this mutation would put the person at a greater risk of breast cancer.

- (b)** DNA sequencing is one method used to test for mutations in *BRCA1*. Before DNA sequencing occurs, the DNA sample goes through the polymerase chain reaction (PCR).

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(c) Genetic screening shows that in the UK:

- 1 in 400 females have a mutated allele of *BRCA1*
- 70% of females with a mutated allele of *BRCA1* will develop breast cancer by the time they are 80 years old.

In 2022, the population of females in the UK was 3.5×10^7 .

Calculate the number of females in the UK population in 2022 who it is estimated will develop breast cancer by the age of 80 years old. Give your answer in standard form.

Show your working.

number of females = [2]

(d) One advantage of genetic screening is to determine which drugs should be selected to treat cancer.

A study was carried out on women with breast cancer who had been treated with a DNA-damaging drug that kills tumour cells.

The women had all been tested for *BRCA1* mutations. One group had a *BRCA1* mutation and the other group did **not** have a *BRCA1* mutation.

The researchers used data to assess the probability of survival for the women in each group of the study after treatment with the drug.



Fig. 5.1 shows the probability of survival for a time period of up to 180 months.

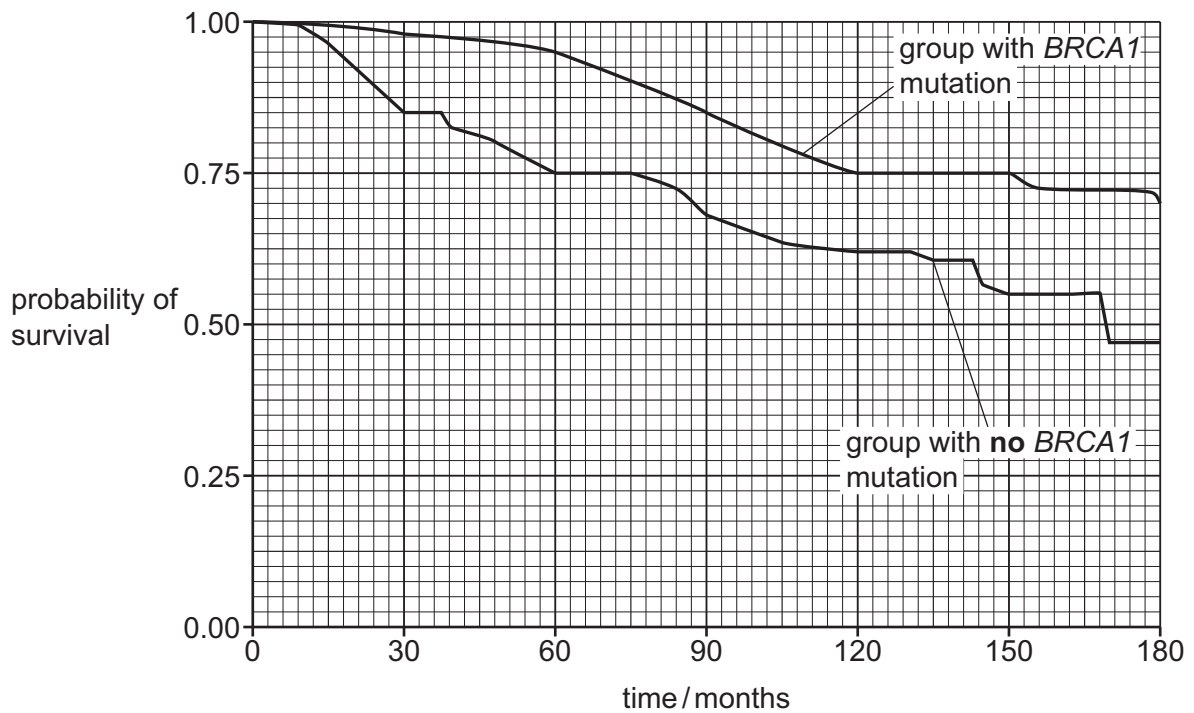


Fig. 5.1

The researchers suggested the hypothesis that the DNA-damaging drug is more effective in women with a *BRCA1* mutation.

Discuss whether the results in Fig. 5.1 support the hypothesis suggested by the researchers.

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[Total: 10]